

The Intelligent System

Software Engineering for Intelligent Systems

JOÃO EDUARDO MONTANDON

What is it?

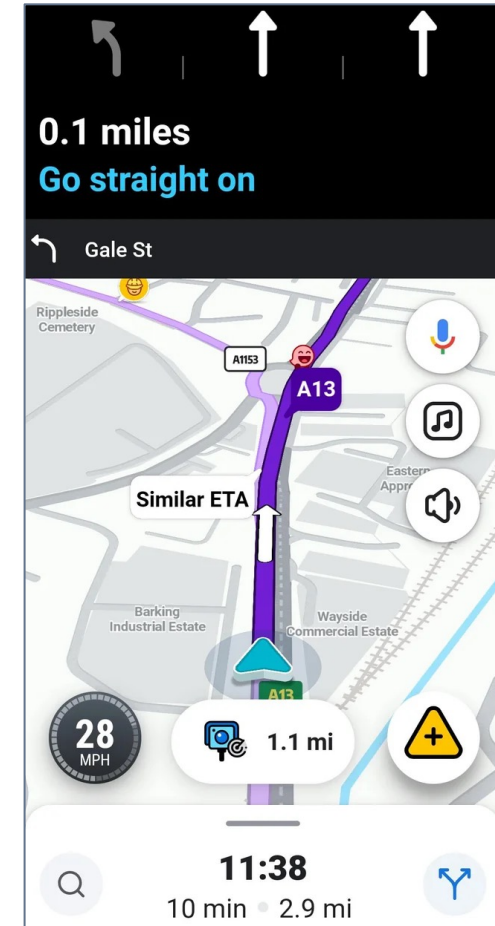
A navigation system

- Calculates the shortest path between two points.
- Provides turn-by-turn navigation



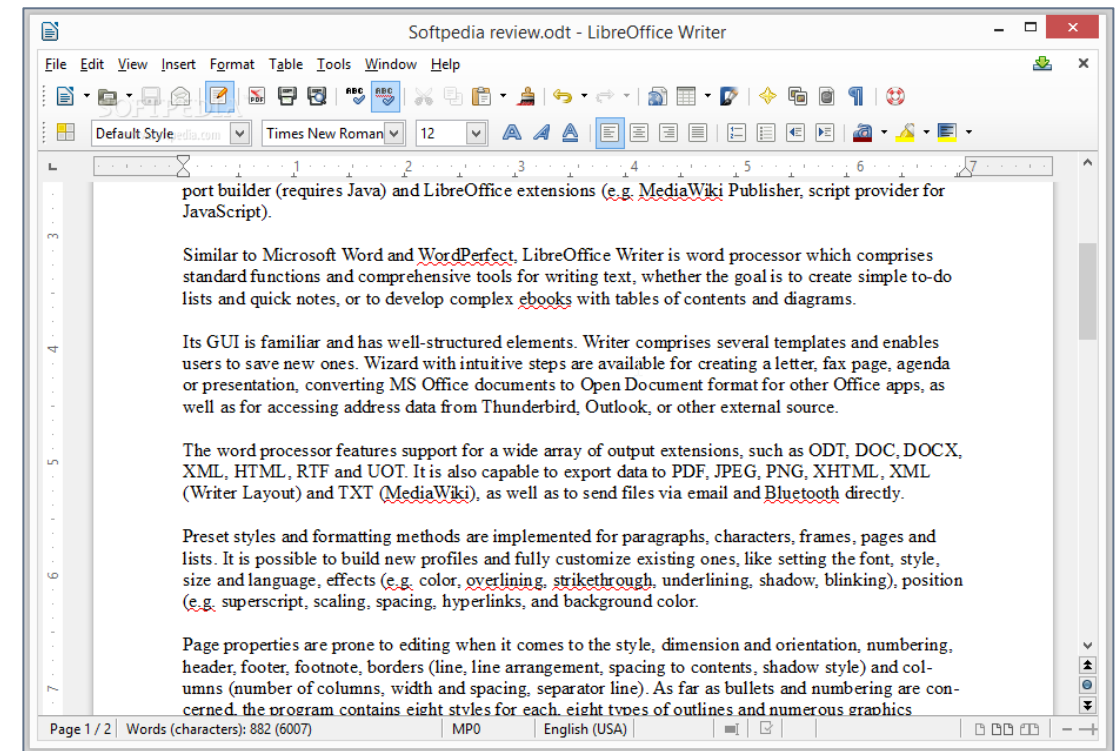
An intelligent navigation system

- ...
- Real-time traffic information.
- Map and routes are dynamically updated.



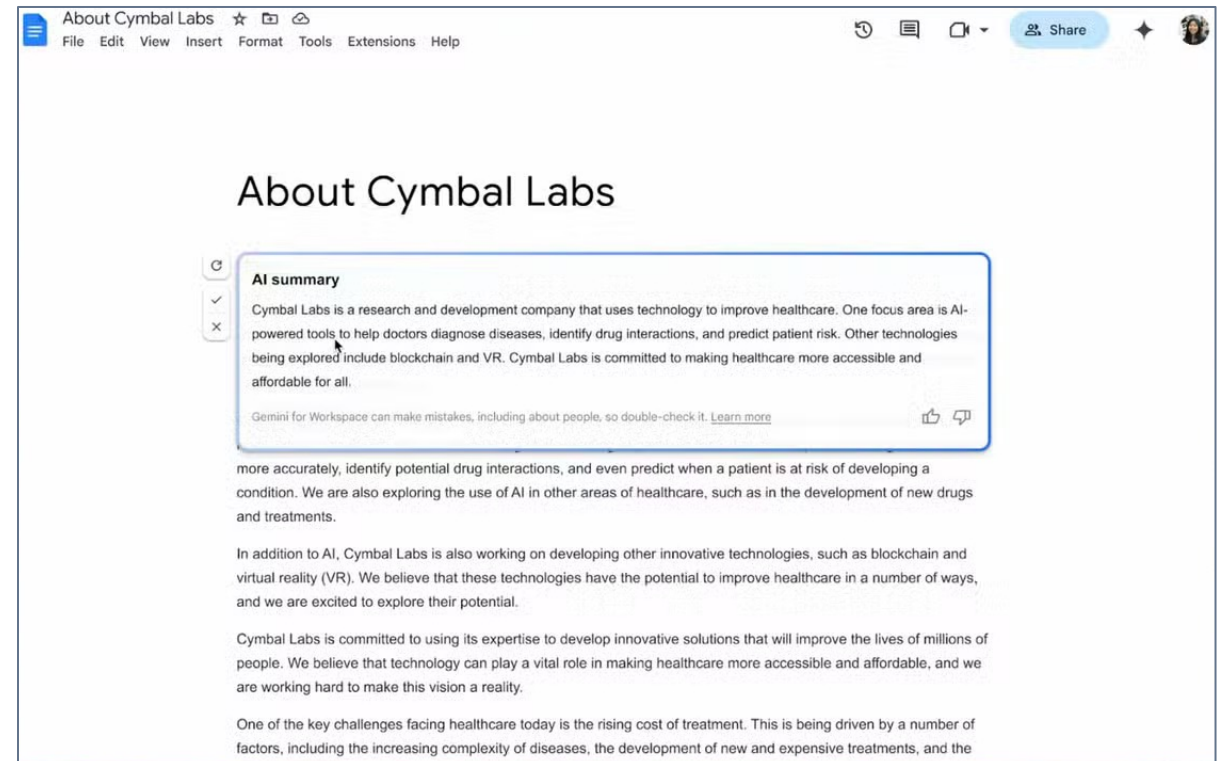
A text editor

- You can write and edit text.
- Can format your text with different styles.
- Performs grammar checking.



An intelligent text editor

- ...
- Summarizes the text.
- Change the text tone.
- Translate the document.

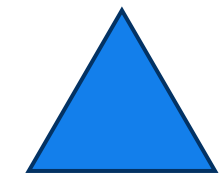


An intelligent system...

*... is one way to **apply machine learning** in practice. It takes intelligence and leverages it **to achieve meaningful objectives**.*

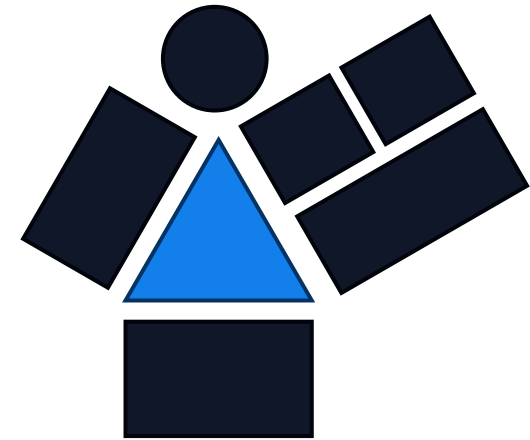
Build an Intelligent System...

is incredibly challenging. Consultants report that **87 percent of machine-learning projects fail** and 53 percent do not make it from prototype to production.



ML model

What makes this so difficult?

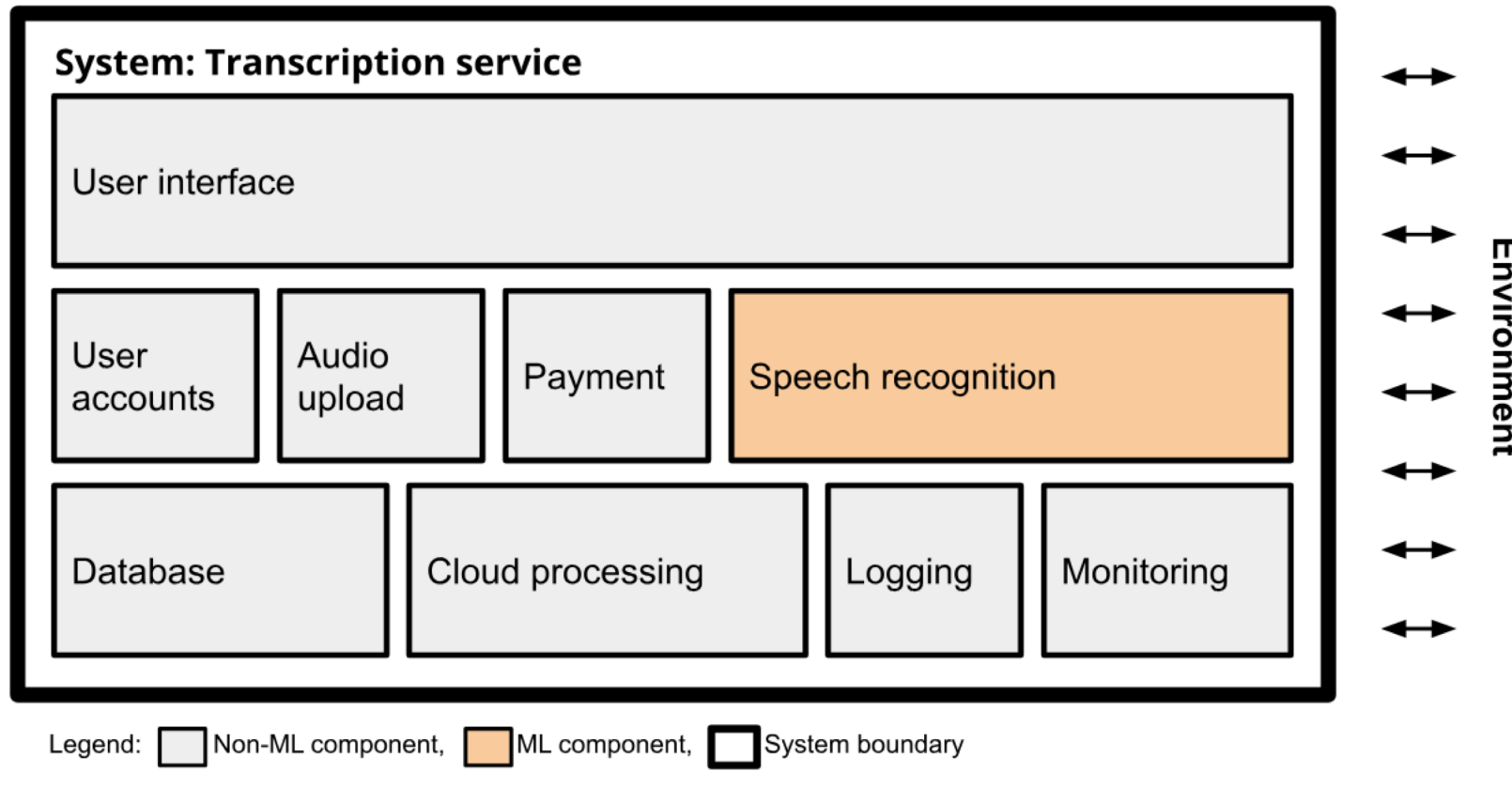


Intelligent System

A Matter of Perspective

System-wide vs Model-centric views

A Typical Intelligent System



Still is a **software system**!

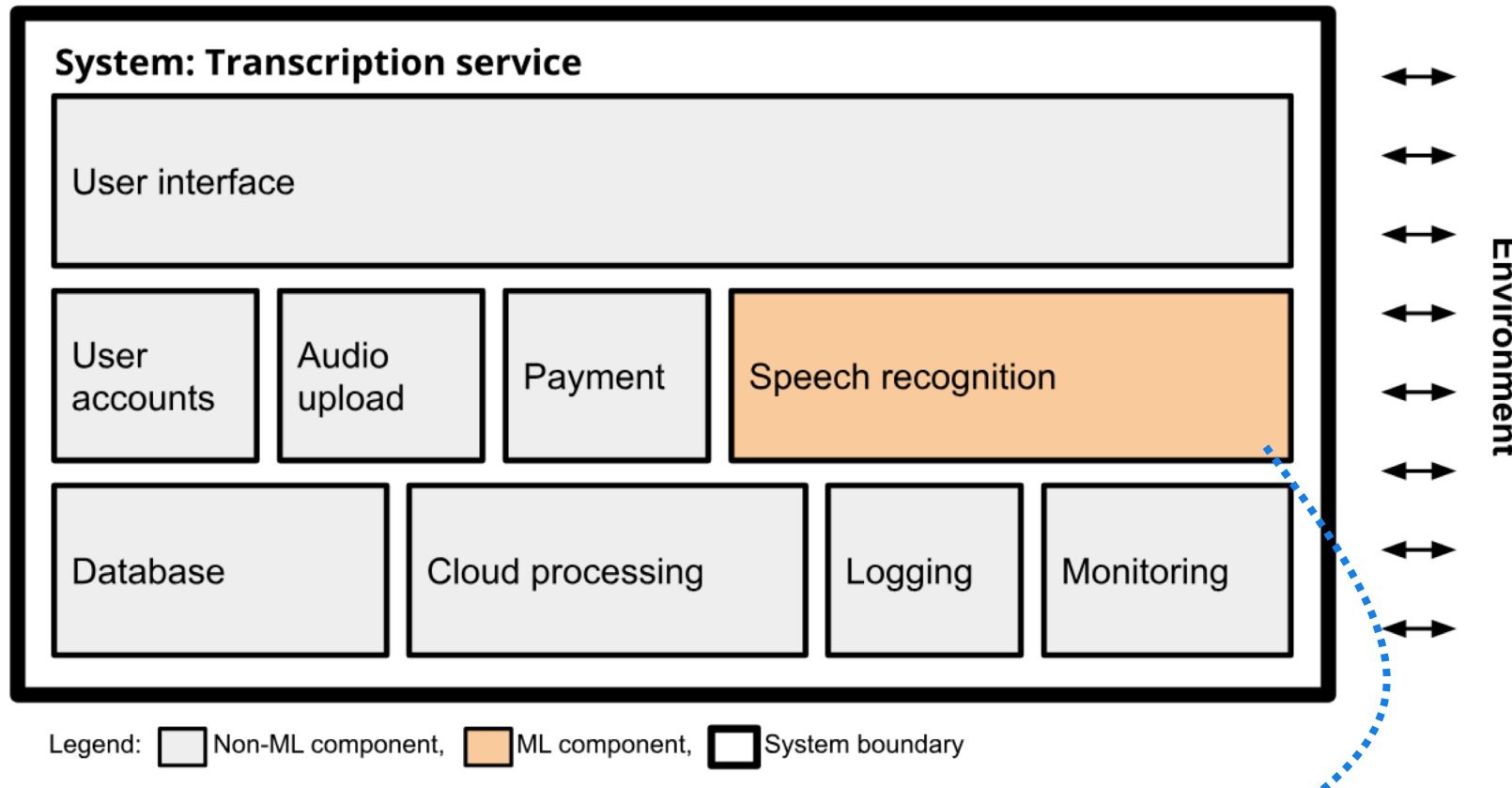
A Typical Intelligent System

ML model is “just” another component of the system.

You still need:

1. A user interface
2. A data storage
3. A monitoring system
4. Communicate with third-party providers
5. ...

A Typical Intelligent System



Have an **atypical component!** 

Inductive Reasoning

Deriving theories
from observations

Strong evidence
suggests a rule

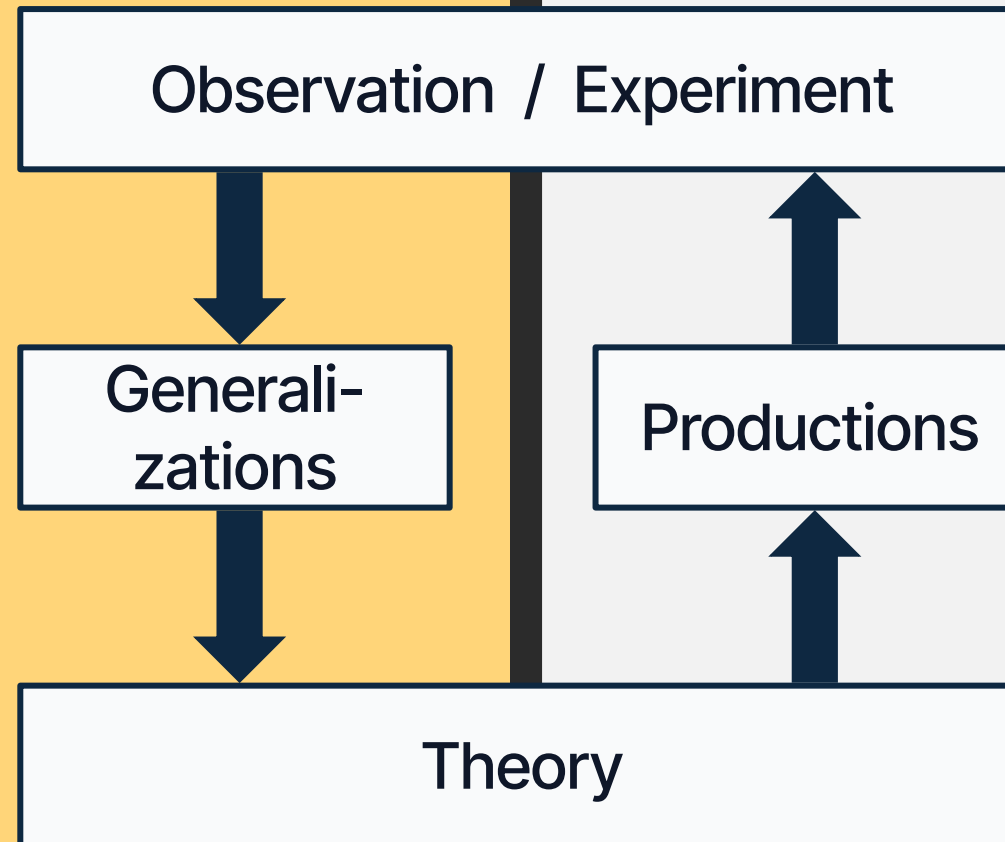
Science-y reasoning

Deductive Reasoning

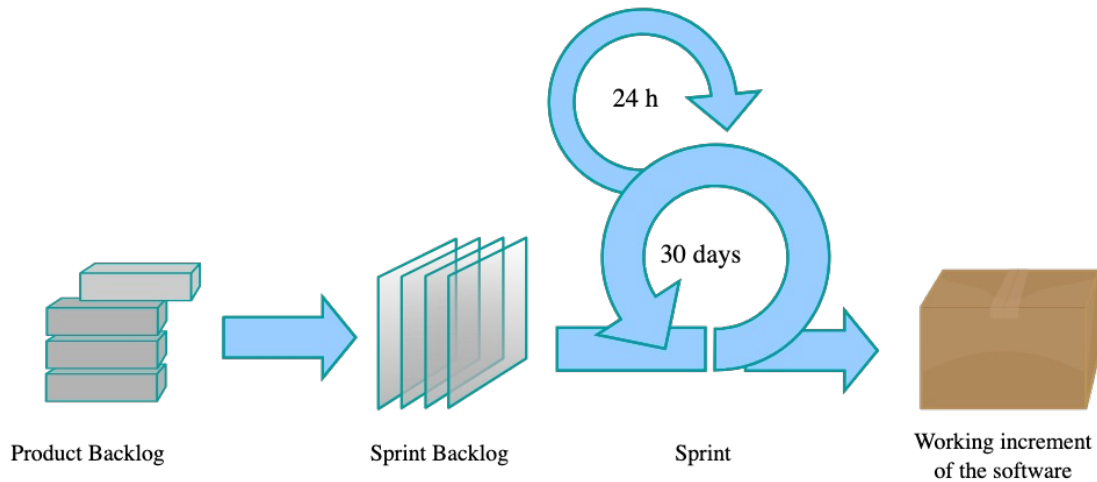
Combining logical
statements

Proving theorems
from axioms

Math-y reasoning

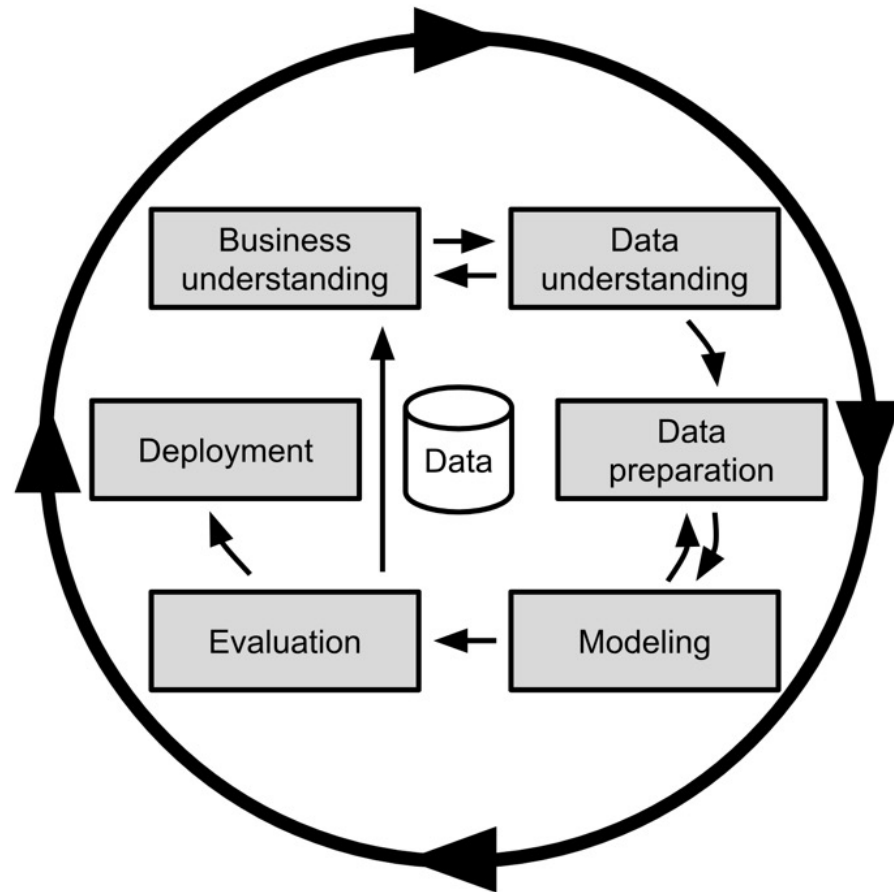


The Software Engineering Process



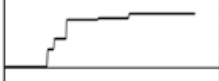
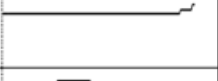
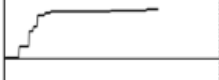
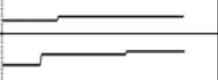
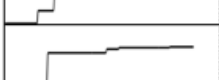
- ❑ Different degrees of iteration.
- ❑ [Major] specifications known upfront.
- ❑ Goal is clearly defined since beginning.
- ❑ Focus on product.

The Data Science Process



- ❑ Is inherently exploratory.
- ❑ Experience guide the process.
- ❑ Constant refinement.
- ❑ Theories emerges from data.
- ❑ Focus on learning.

The Data Science Process

	First 2 Hours	Second 2 Hours	Final Accuracy
TAP1			84.7%
TAP2	X	X	75.3%
TAP3			78.3%
TAP4			82.9%
TAP5			84.7%
TAP6			78.0%
TAP7			56.9%
TAP8			22.8%
TAP9			78.8%
TAP10			84.4%

Progress is not granted!

Nearly impossible to determine how long it will take.

The Data Scientist Toolbelt

Idea from *literate programming*, where code and text are interleaved to form a narrative (Knuth, 80s).

First product: Wolfram Mathematica 1.0 (1988).

Sequence of text and code cells. **Code can be executed** per cell in a single environment (kernel). **Text explains** and show execution results.

Many languages available (R, Julia, Ruby, JavaScript, **Python**, ...)

```
In [1]: import matplotlib.pyplot as plt
import numpy as np
```

Code cells show both code input and output:

```
In [2]: 6*8 - 3*2
```

```
Out[2]: 42
```

Markdown cells such as this one can contain text and $LAT_{E}X$ equations such as $c(a,b) = \sqrt{a^2 + b^2}$. We can use code to define the corresponding functions:

```
In [3]: def c(a, b):
return (a**2 + b**2)**0.5
```

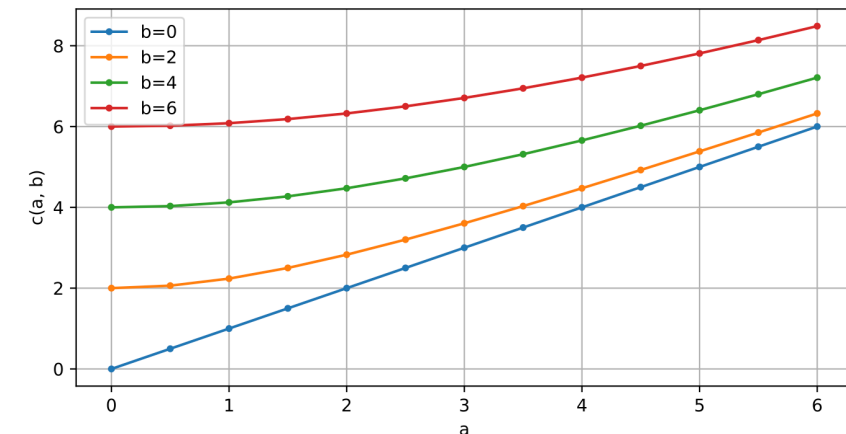
Let us compute $c(a, b)$ as a function of a and b and plot multiple lines (each for a fixed b).

```
In [4]: a = np.linspace(0, 6, 13)

plt.figure(figsize=(8, 4))

for b in [0, 2, 4, 6]:
    c_res = c(a, b)
    plt.plot(a, c_res, '.-', label='b='+str(b))

plt.xlabel('a')
plt.ylabel('c(a, b)')
plt.grid()
plt.legend();
```



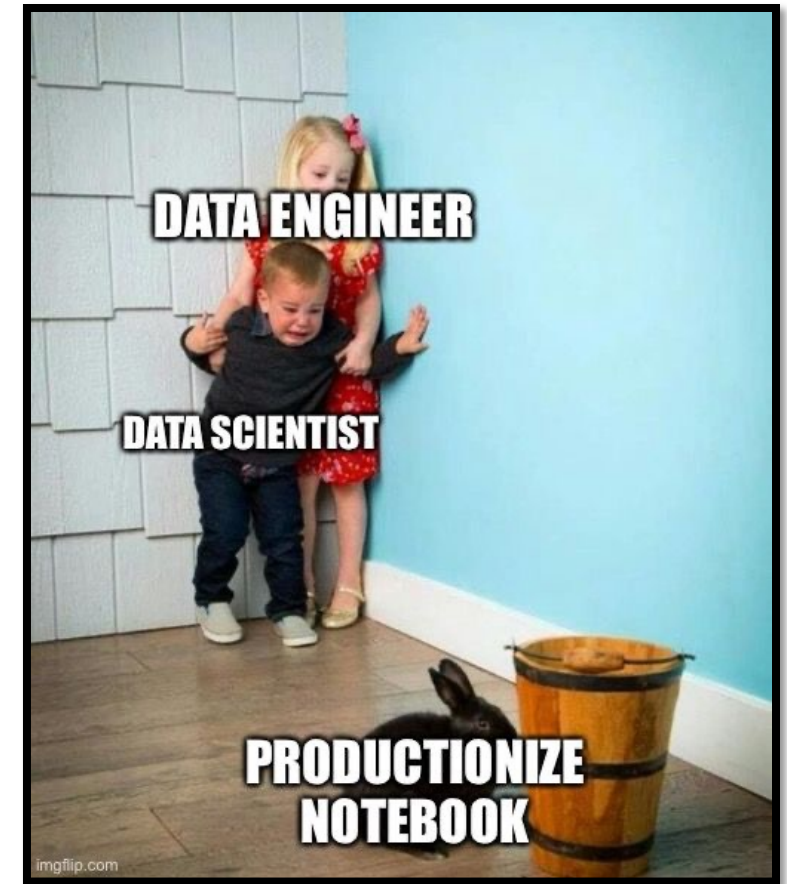
Markdown cells can be used to comment on and interpret the results.

The Data Scientist Toolbelt

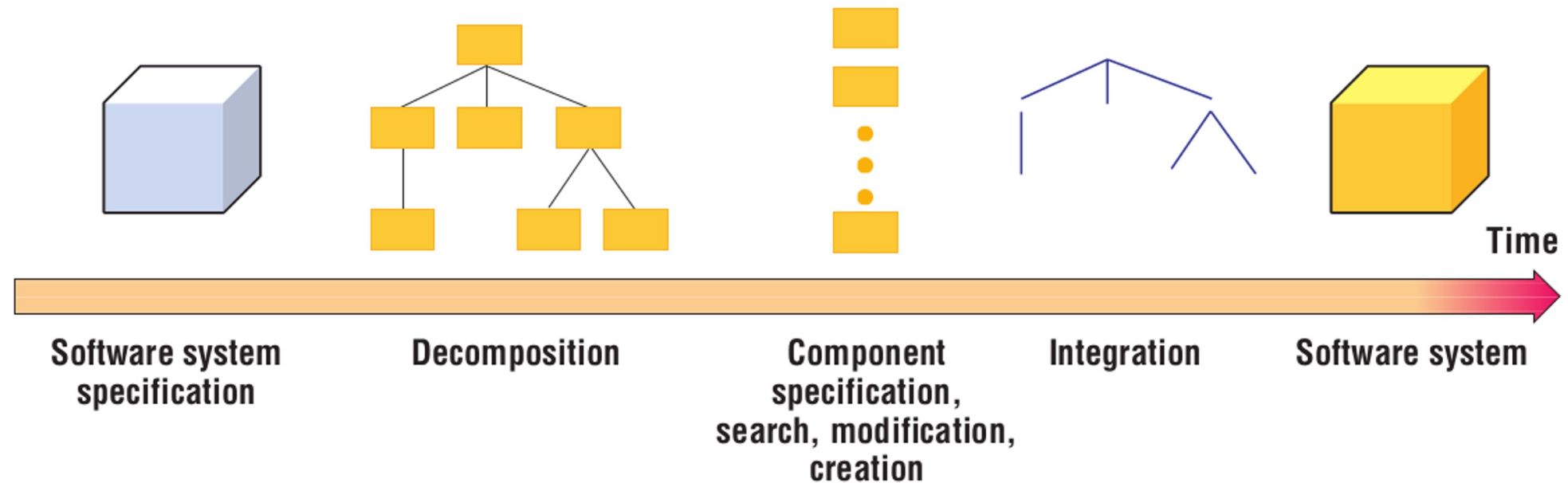
- **Excellent tool for exploration and prototyping.**
- Quick and visual feedback (e.g., tables and figures).
- Enables partial and incremental computation (per cell execution).
- Quick and easy to use.
- Easy to integrate computation results with documentation (e.g., export report to pdf/html).

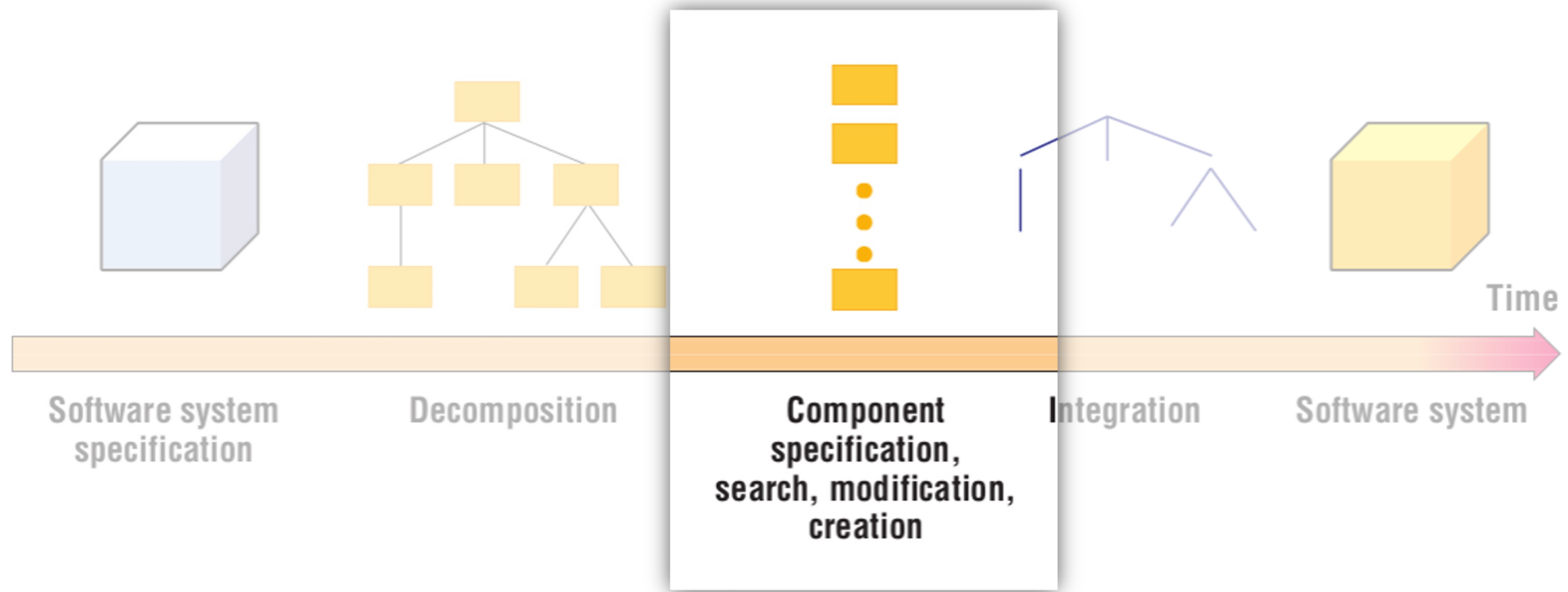
The Data Scientist Toolbelt

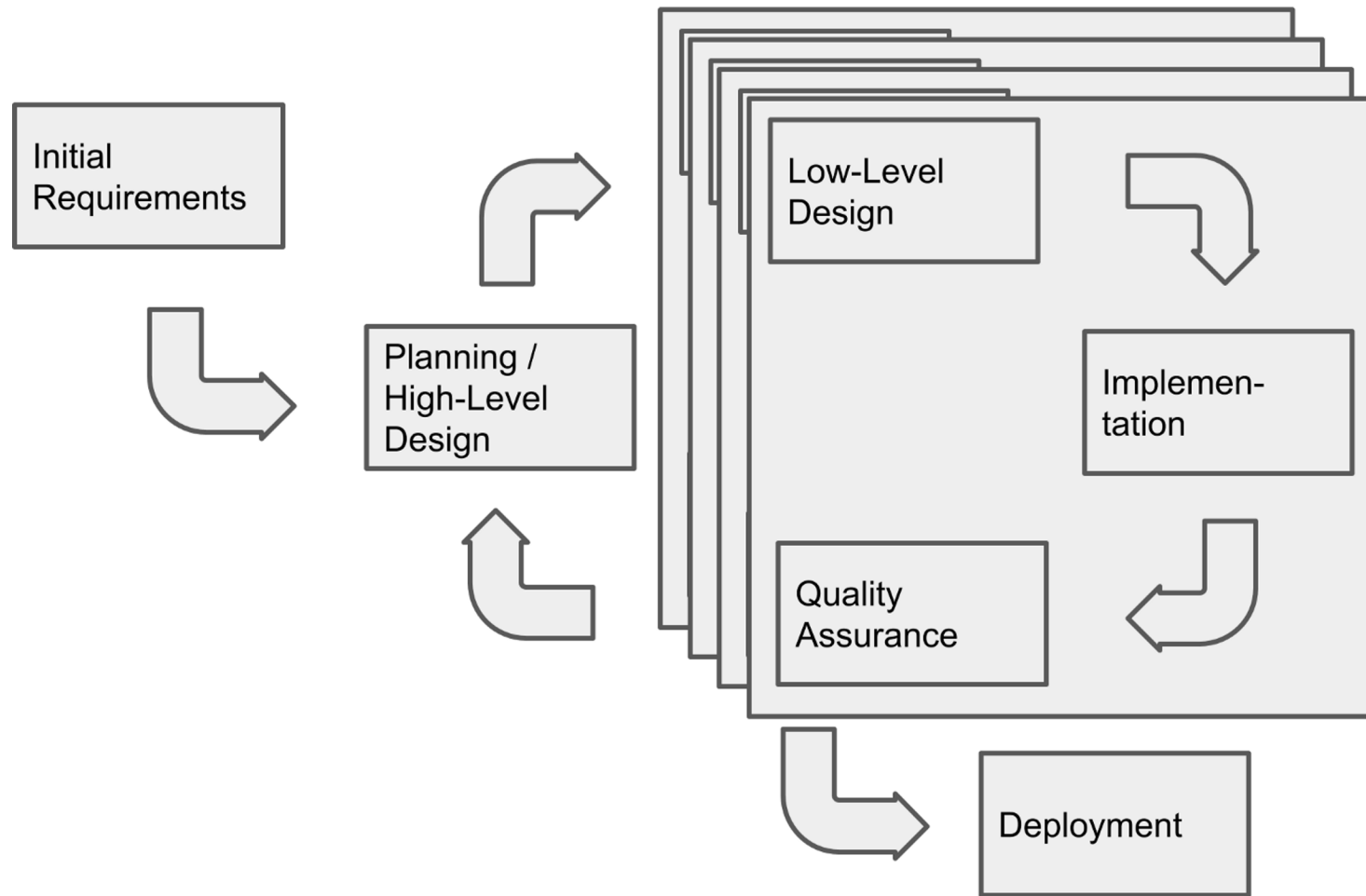
- **Terrible for shipping a product!**
- Out-of-order execution.
- lack of abstraction.
- lack of automated testing.
- global variables.
- Poor version control.
- ...



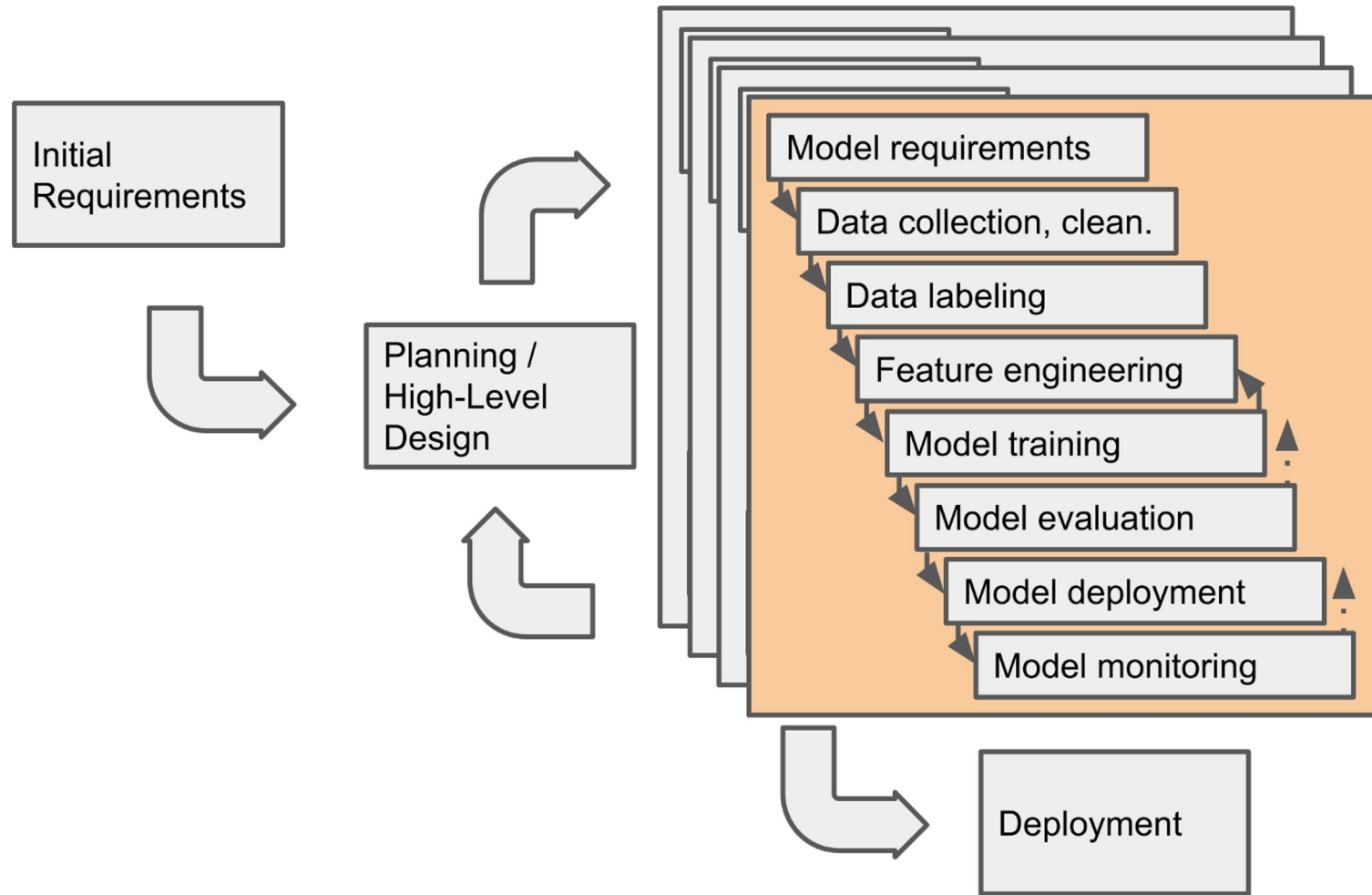
Software Development: Divide and Conquer





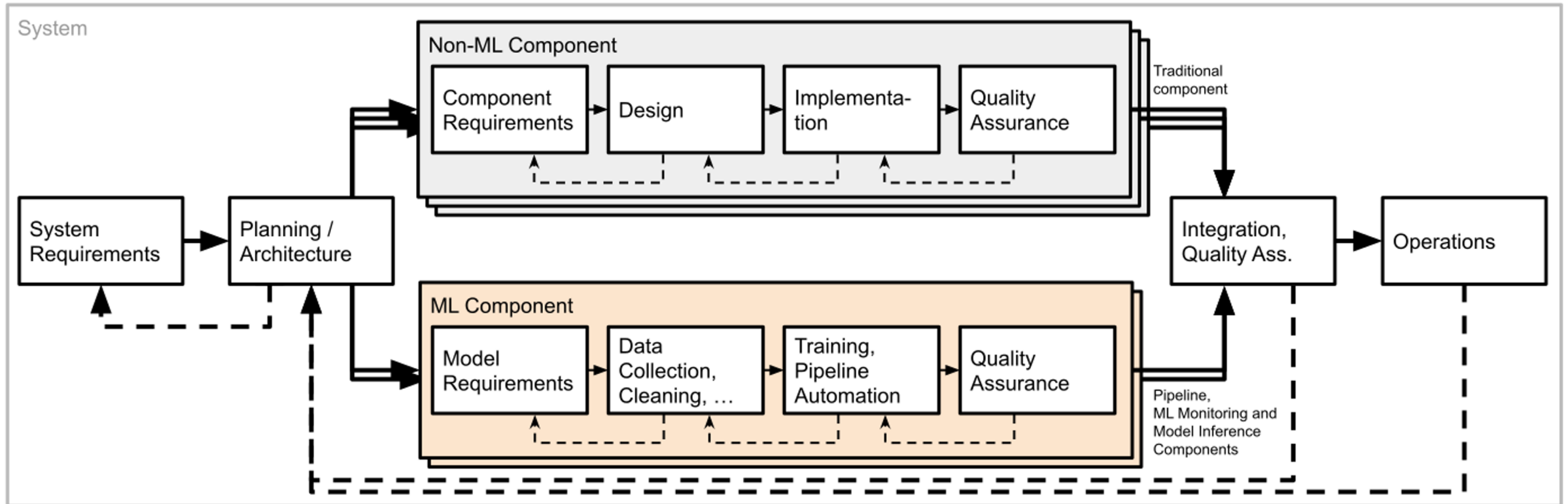


A component dev. cycle



An **intelligent** component dev. cycle

An Intelligent System Dev. Cycle



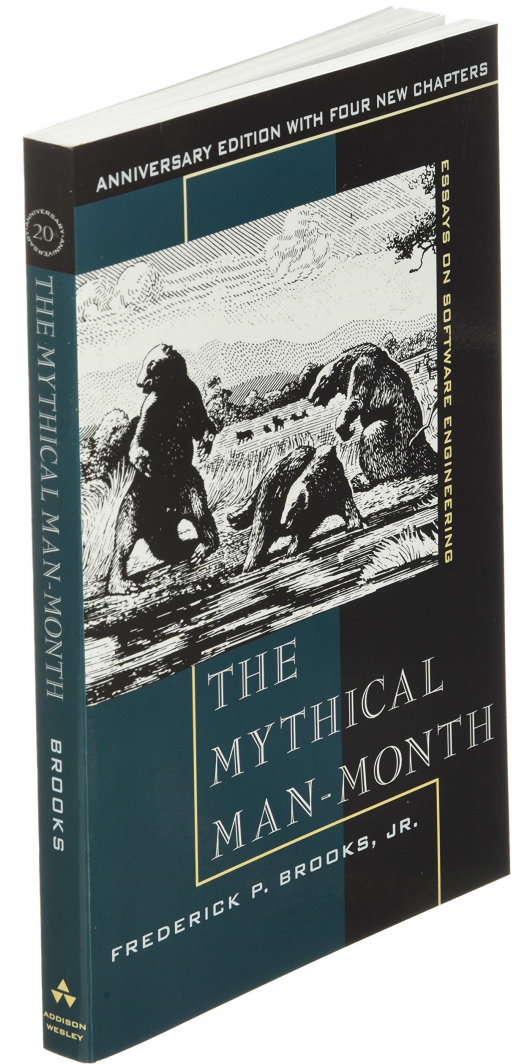
Must be Touched by More Hands

Interdisciplinary Teams

The Surgical Team

*Mills proposes that each segment of a large job be tackled by a team, but that the **team be organized like a surgical team** [...]*

-- Fred Brooks



The T-Shaped Professional



Generalist

Broad knowledge of
other topics

+



I-Shaped

Deep expertise in
one topic

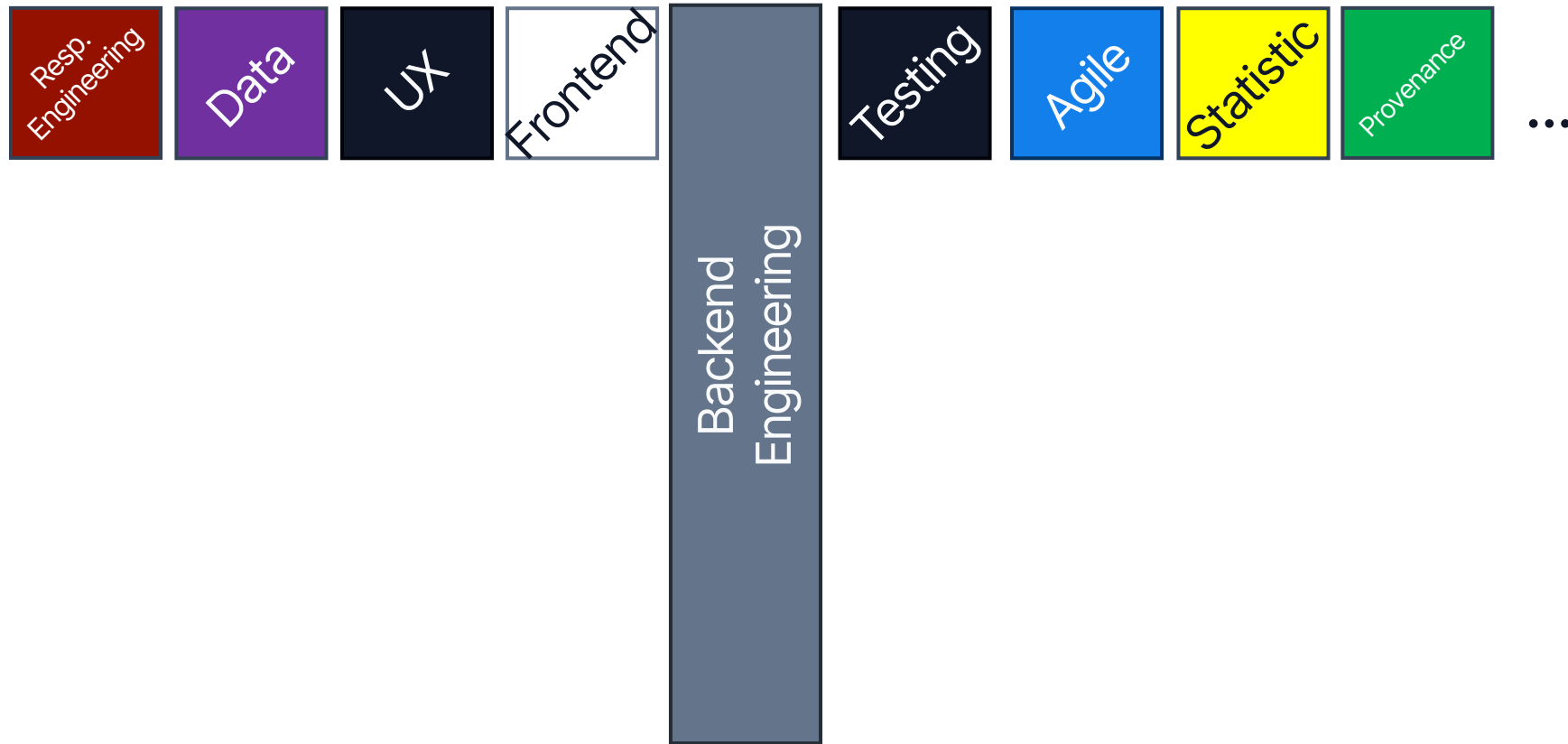
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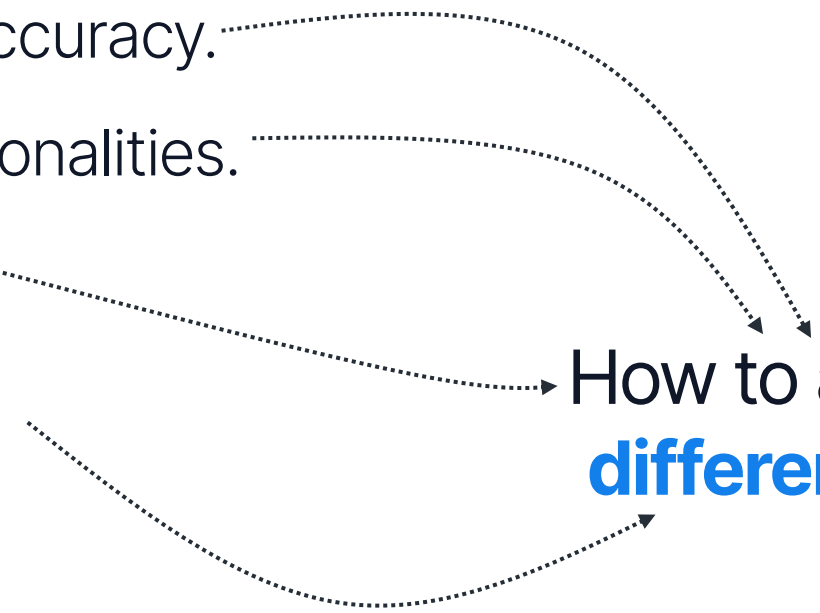
T-Shaped

Expert in one topic, broad
knowledge in others

Interdisciplinary T-Shape

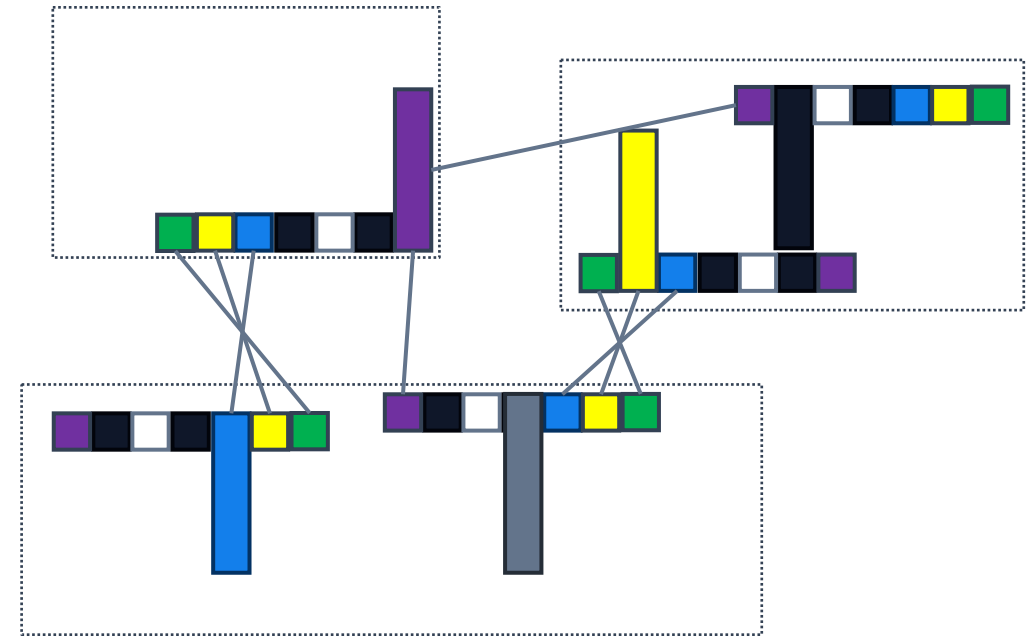


Conflicts in Interdisciplinary Teams

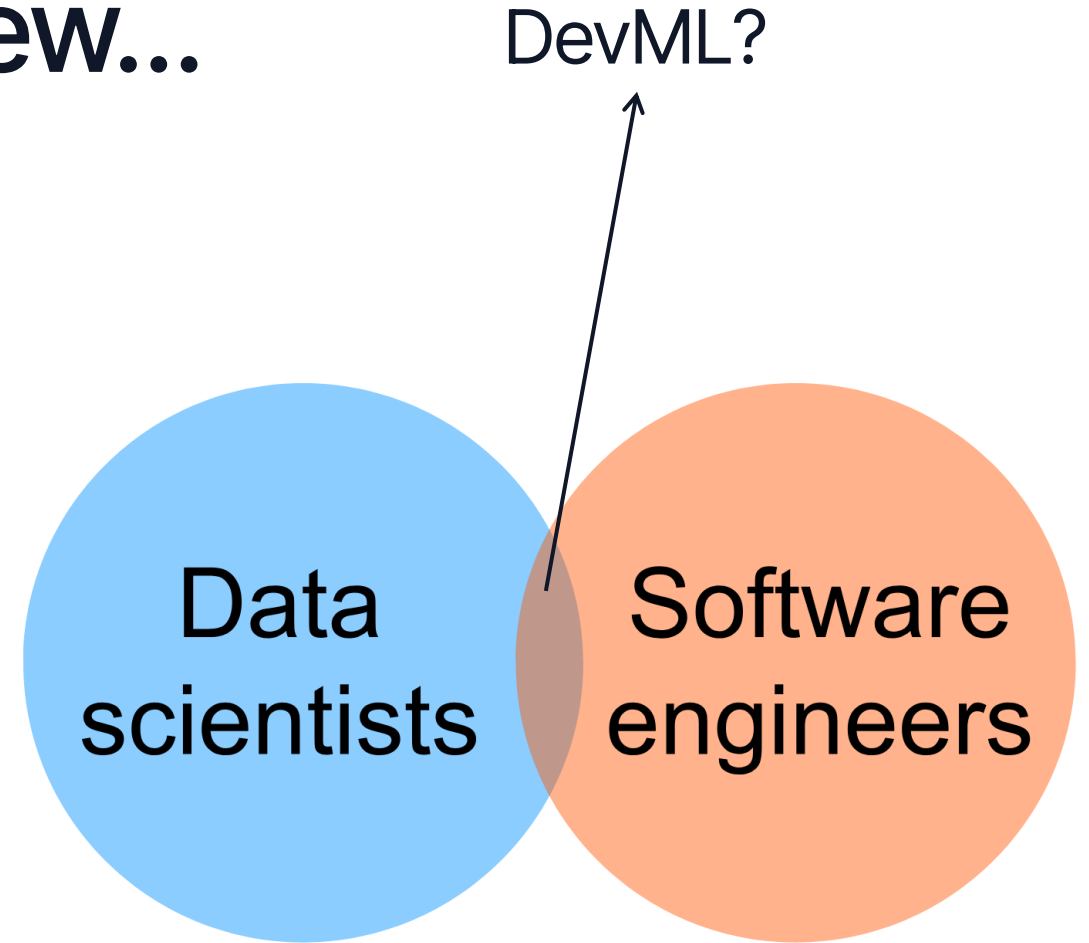
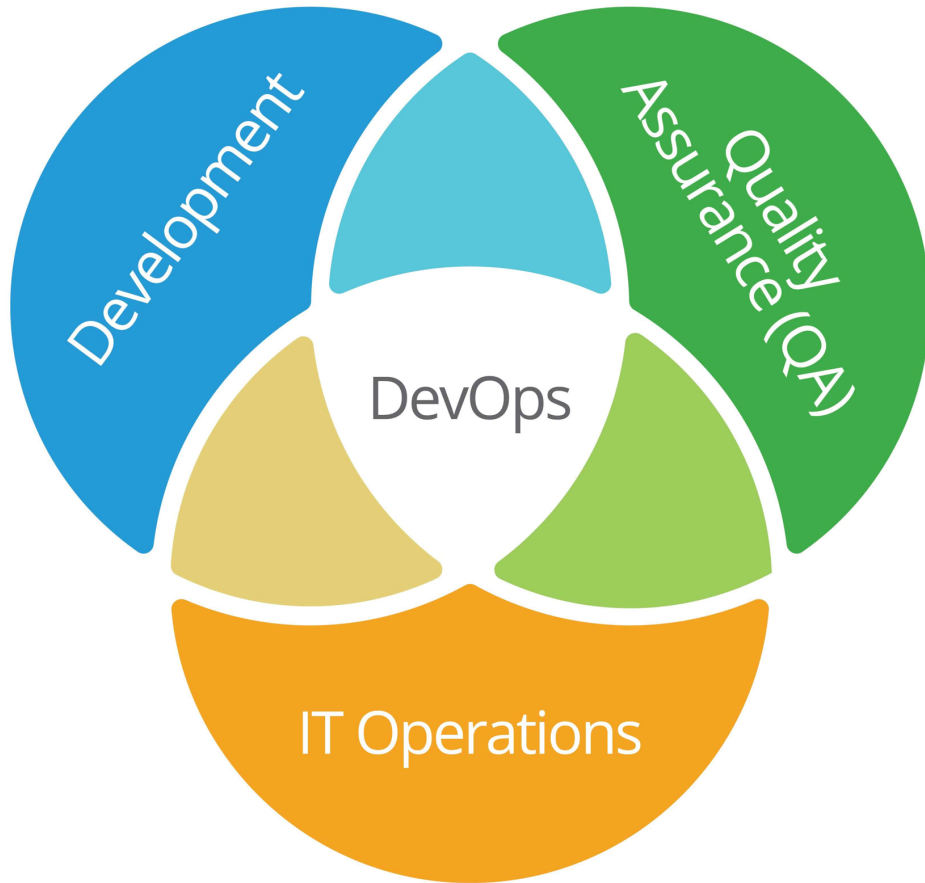
- Intelligent Systems involve an even diverse background.
 - Each specialist has its own purpose.
 - Data Science → achieve high accuracy.
 - Developers → Implement functionalities.
 - Data Engineers → Collect data.
 - Lawyers → Verify legal aspects.
 - ...
- How to align such **different goals?**
- 
- A diagram consisting of five dotted arrows. Each arrow originates from a goal statement in the bulleted list and points towards the central question 'How to align such different goals?'. The arrows are: 1. From 'Data Science → achieve high accuracy.' to the question. 2. From 'Developers → Implement functionalities.' to the question. 3. From 'Data Engineers → Collect data.' to the question. 4. From 'Lawyers → Verify legal aspects.' to the question. 5. From '...' to the question.

Conflicts in Interdisciplinary Teams

- Communication helps align individual goals into product development.
- Horizontal knowledge to enable communication.
- Deep knowledge to effectively work on the project



This challenge is not new...

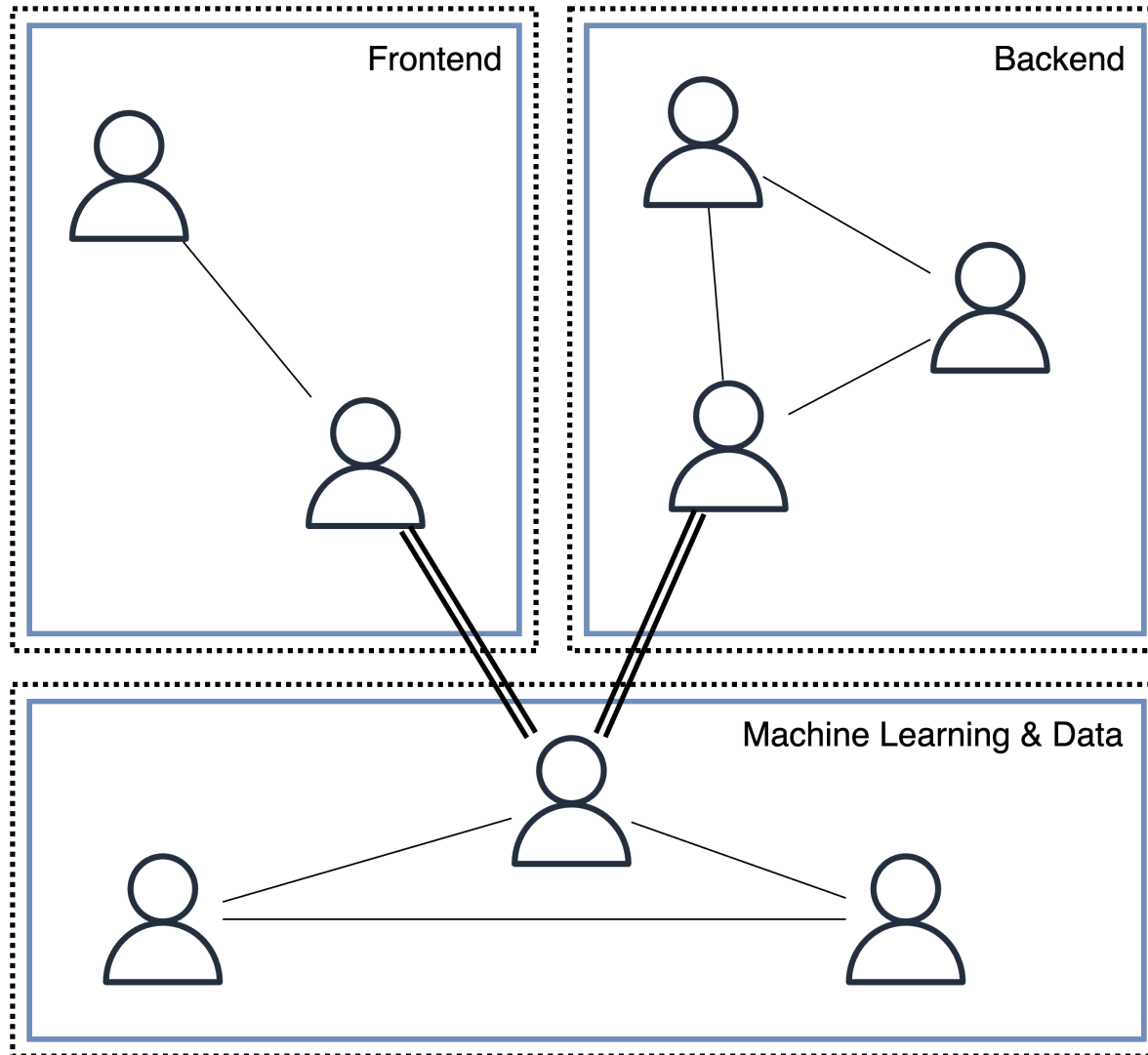


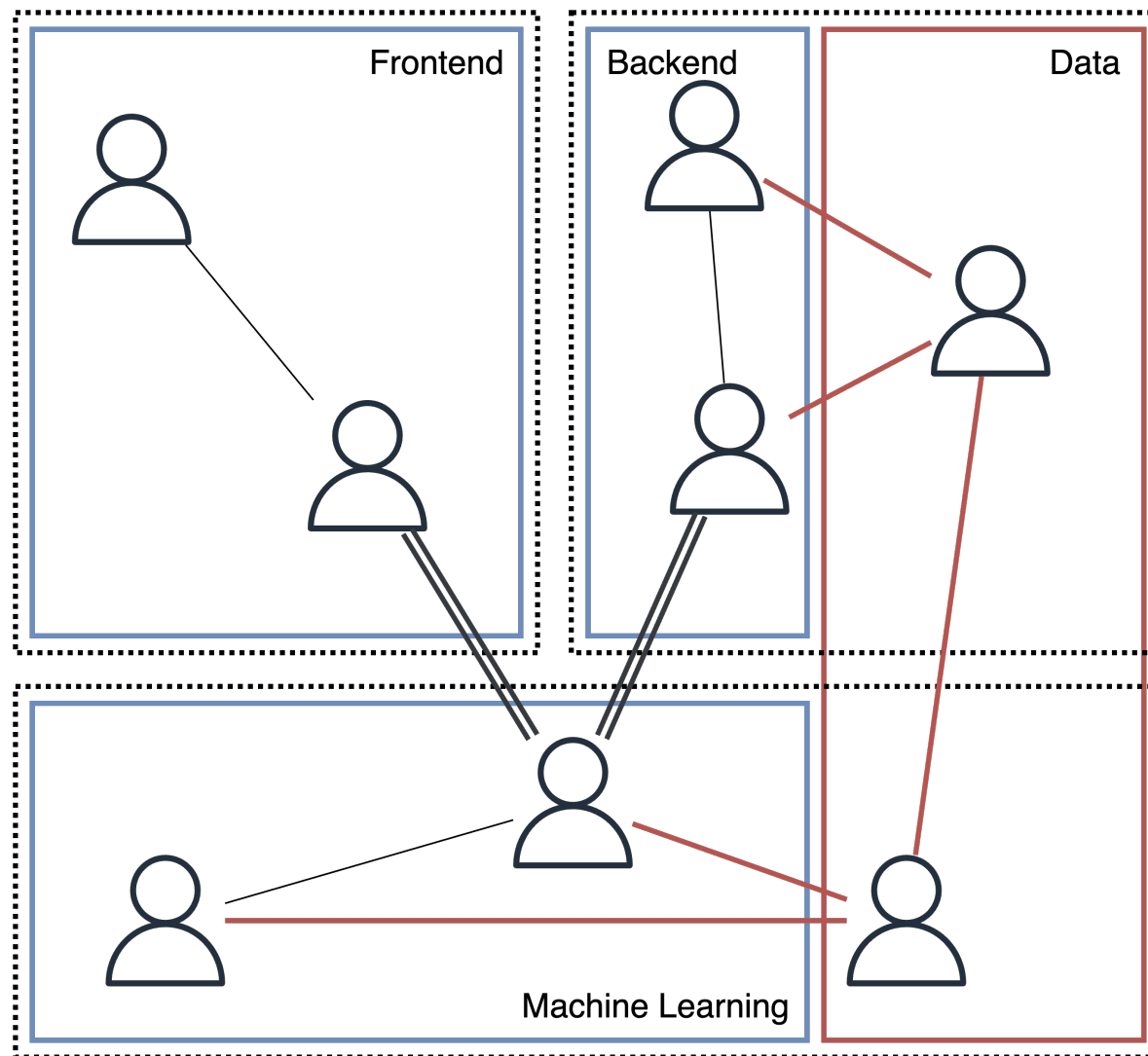
Coordination Costs

*If you have **three groups** working on a compiler, you'll get a **3-pass** compiler; if you have **four groups**, you'll get a **4-pass** compiler*

-- Eric Raymond, about Conway's Law

Interfaces = Information Hiding!



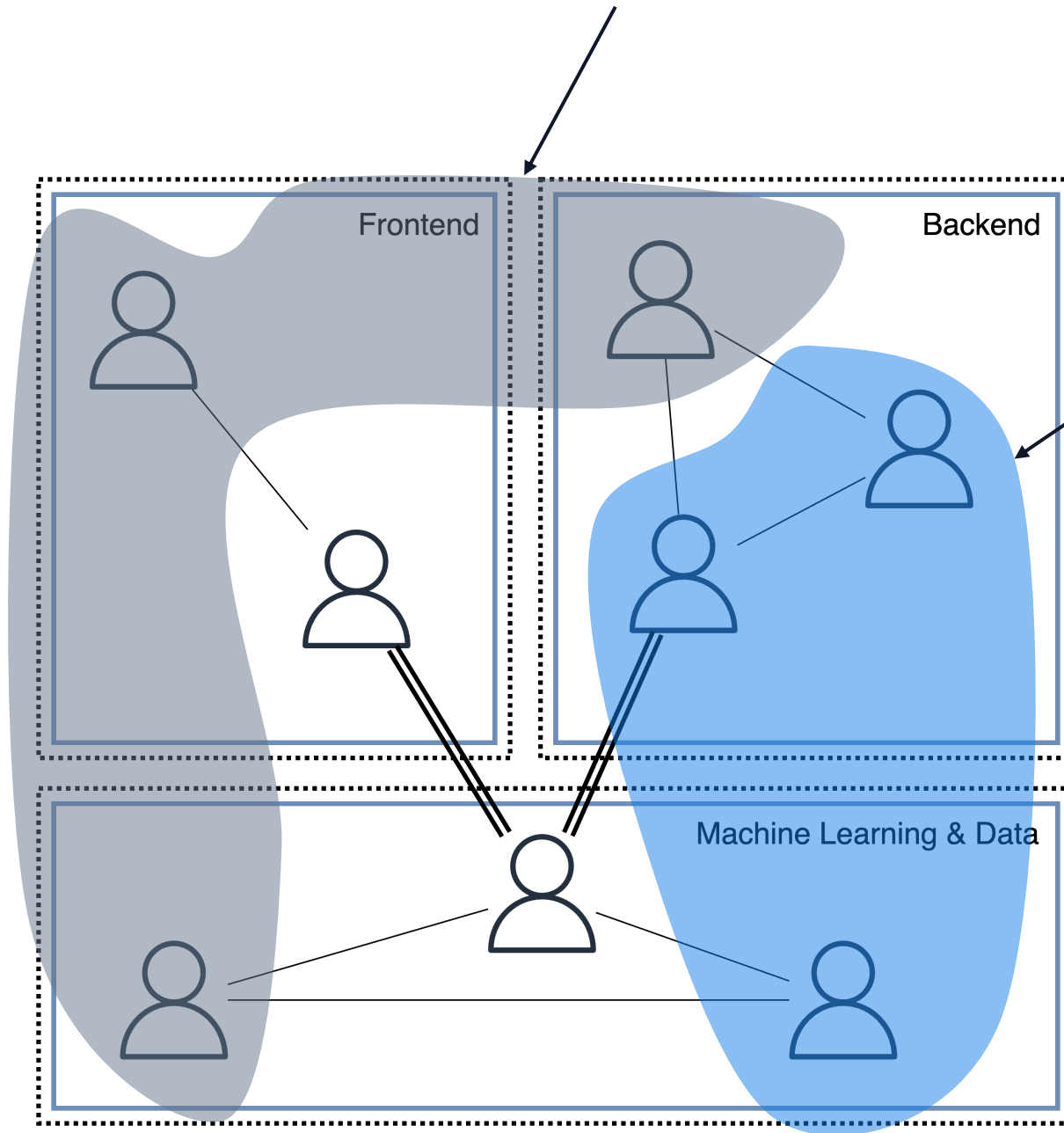


#1: Interfaces are rarely stable

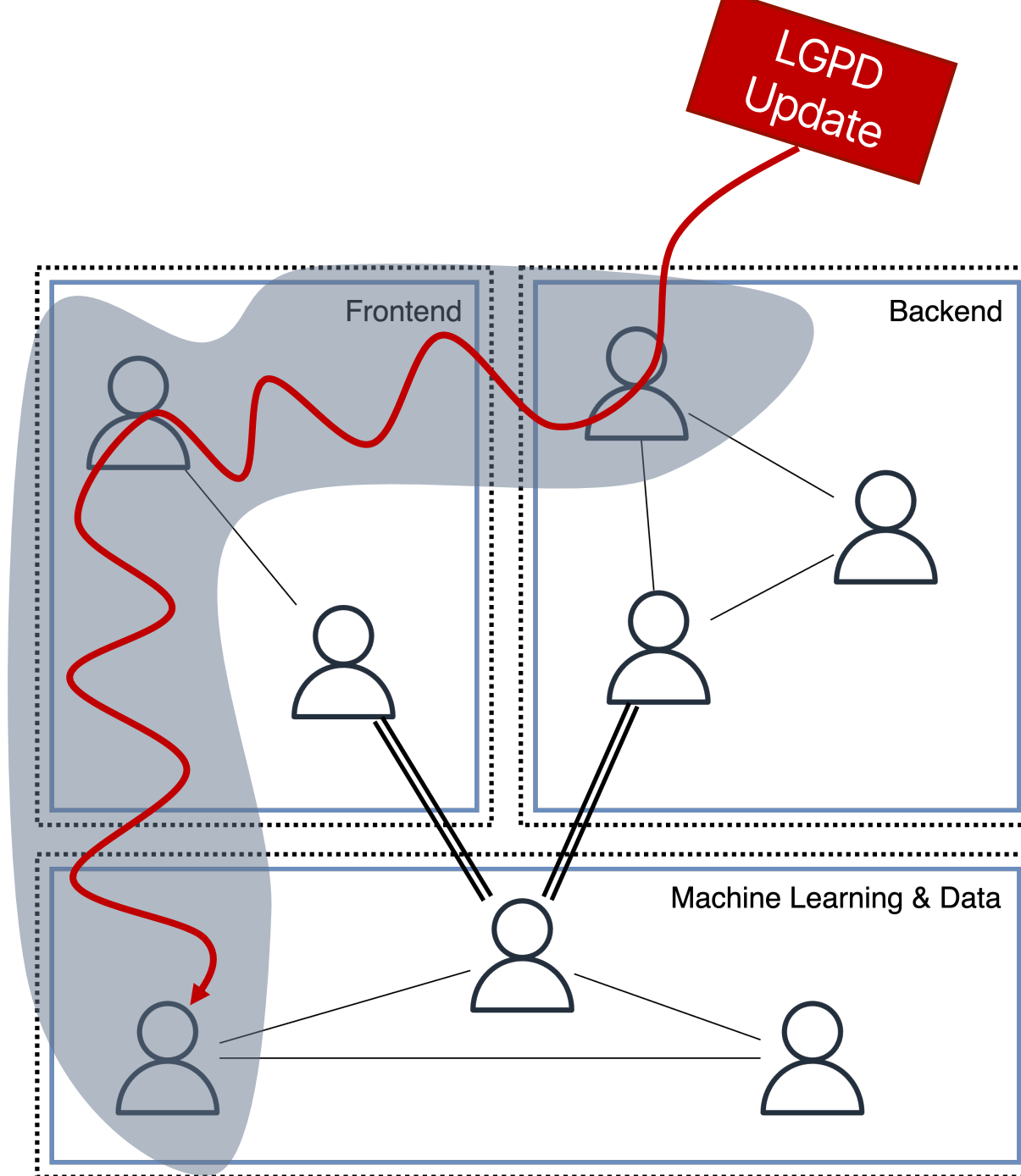
Ethics

Fairness

#2: Some qualities are cross-cutting



Awareness



If information can not be hidden,
broadcast their changes, instead!

Examples: announcement in slack,
notifications in GitHub issues, etc.

Other Challenges

- How to support regression?
- How to map what it should do?
- How to monitor its performance?
- Can I version it?
- How to ensure fairness and ethics?
- Which securities measures?

Other Challenges

- How to support regression? **Evaluation**
- How to map what it should do? **Requirements**
- How to monitor its performance? **Operations**
- Can I version it? **Provenance**
- How to ensure fairness and ethics? **Responsibility**
- Which securities measures? **Security & Safety**



SE4AI

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Obrigado!